

HYBRID_V3

Phase 4 — Dynamic Model Selection

Technical Progress & Experimental Results Report

Scope: Prediction-feature extraction, reliability-based dynamic selection, pairwise selector learning, corrected evaluation, and current conclusions.

1. Objective

Phase 4 aims to dynamically select the model most likely to be correct for each individual ISIC 2019 image. The hypothesis is that the six models have complementary error patterns, so sample-dependent selection can outperform fixed models and simple fusion.

2. Model Pool

Model	Family
ResNet50	CNN
EfficientNet-V2-S	CNN
EfficientNet-B3	CNN
MobileNet-V3-Large	CNN
DeiT-Tiny	ViT
Swin-Tiny	ViT

3. Dataset & Evaluation

- ISIC 2019 dataset: 25,331 images.
- Leakage-safe split: 20,053 train, 4,011 validation, 1,267 test.
- Phase 4 prediction features: 1,267 samples.
- Latest corrected pairwise evaluation: 254 samples, seed 42.
- Classes: AK, BCC, BKL, DF, MEL, NV, SCC, VASC.

4. Prediction Features

Features represented each model's prediction behavior and cross-model agreement.

Group	Features
Prediction	Predicted class per model
Confidence	Confidence, max/min/mean/std, top/second confidence
Uncertainty	Entropy statistics
Margin	Per-model prediction margin
Agreement	Unique predictions, majority count, agreement ratio
Family	CNN–ViT agreement
Relations	Confidence gap and spread

Data-quality issue: all six margin columns were missing in the source feature table, so scikit-learn skipped them during imputation. They should be regenerated before the final system.

5. Base Model Performance

Model	Accuracy	Correct
ResNet50	0.7388	936/1267
EfficientNet-V2-S	0.7443	943/1267
EfficientNet-B3	0.7096	899/1267
MobileNet-V3-Large	0.7364	933/1267
DeiT-Tiny	0.7214	914/1267
Swin-Tiny	0.7301	925/1267

6. Initial Reliability Selectors

A separate selector was trained for each model to estimate whether that model would be correct for a given sample.

Selector	Accuracy	Bal. Acc.	ROC-AUC
ResNet50	0.7520	0.7636	0.8360
EfficientNet-V2-S	0.7913	0.7891	0.8462
EfficientNet-B3	0.7874	0.7824	0.8577
MobileNet-V3-Large	0.7598	0.7842	0.8318
DeiT-Tiny	0.8110	0.8128	0.8620
Swin-Tiny	0.7717	0.7569	0.8167

Initial selection comparison:

Strategy	Accuracy
Majority vote	0.8307
Confidence winner	0.7992
Dynamic selector	0.8150
Oracle	0.9291

The initial dynamic selector reached 81.50%, but majority voting was stronger at 83.07%. The 92.91% oracle established substantial headroom.

7. Pairwise Selection

To simplify the six-way competence problem, Phase 4 introduced pairwise selectors. Six models create 15 unique pairs. Each pair learns which model is more likely to be correct using model identity, prediction, confidence, and confidence-difference features.

- Pairwise dataset: 3,642 rows.
- Target distribution: 1,875 vs 1,767.
- 15/15 model pairs successfully trained.
- Mean training accuracy: 0.6514.
- Mean balanced accuracy: 0.6483.
- Mean ROC-AUC: 0.7059.
- Best training ROC-AUC: 0.8076 for ResNet50 vs EfficientNet-B3.

8. Pairwise Training Results

Pair	Acc.	Bal.	AUC
DeiT vs Swin	0.6327	0.6313	0.6957
Eff-B3 vs DeiT	0.7222	0.7166	0.7145
Eff-B3 vs MobileNet	0.6800	0.6753	0.7175
Eff-B3 vs Swin	0.6250	0.6190	0.6490
Eff-V2-S vs DeiT	0.6250	0.6402	0.7478
Eff-V2-S vs Eff-B3	0.7292	0.7179	0.7821
Eff-V2-S vs MobileNet	0.7317	0.7321	0.7416
Eff-V2-S vs Swin	0.6667	0.6590	0.6842
MobileNet vs DeiT	0.6042	0.6031	0.6993

Pair	Acc.	Bal.	AUC
MobileNet vs Swin	0.5652	0.5606	0.6117
ResNet vs DeiT	0.5536	0.5372	0.6256
ResNet vs Eff-B3	0.7115	0.7136	0.8076
ResNet vs Eff-V2-S	0.6735	0.6717	0.7700
ResNet vs MobileNet	0.6538	0.6538	0.6849
ResNet vs Swin	0.5962	0.5933	0.6563

9. Corrected Pairwise Dynamic Evaluation

The evaluation pipeline encountered sample-ID/class-alignment issues in earlier versions. These were corrected. The latest run includes a sanity check of true classes and predicted labels before computing the final metrics. This corrected run is the current reference benchmark.

Strategy	Accuracy	Bal. Acc.	Macro F1
Oracle	0.9134	0.8349	0.7832
Pairwise vote	0.7874	0.5886	0.5465
Pairwise probability	0.7756	0.5623	0.5305
Prior-corrected pairwise	0.7717	0.5589	0.5285
EfficientNet-V2-S	0.7559	0.5812	0.5283
Majority vote	0.7520	0.5798	0.5102
Confidence winner	0.7520	0.7196	0.5899
ResNet50	0.7441	0.5787	0.5144
DeiT-Tiny	0.7362	0.6443	0.5425
MobileNet-V3-Large	0.7323	0.4966	0.4405
Swin-Tiny	0.7047	0.5570	0.4569
EfficientNet-B3	0.6850	0.6153	0.4869

10. Interpretation

- **Best actual strategy:** pairwise vote, 78.74% accuracy.
- **Oracle:** 91.34%, leaving a 12.60 percentage-point oracle gap.
- **Best fixed model:** EfficientNet-V2-S, 75.59%.
- **Best balanced accuracy among actual strategies:** confidence winner, 71.96%.
- Pairwise selection currently improves overall accuracy over all fixed models in the corrected evaluation.
- Accuracy, balanced accuracy, and macro F1 disagree in ranking because the dataset is highly class-imbalanced.
- The earlier 86.15% pairwise-vote result should not be used as the final benchmark because later corrections identified evaluation/alignment issues.

11. Current Phase 4 Findings

- The model pool contains meaningful complementary errors.
- The oracle result confirms that dynamic selection has substantial theoretical headroom.
- Majority voting is a strong and necessary baseline.
- The first reliability selector did not beat majority voting.
- Pairwise competence learning is promising, but aggregation is not yet optimal.
- Missing margin features are an important upstream data-quality issue.
- Class-aware metrics must remain part of all future evaluations.

12. Recommended Next Experiments

- Fix and regenerate all missing margin features.
- Use repeated stratified splits or cross-validation instead of one 254-sample split.

- Calibrate each base model's confidence before pairwise learning.
- Learn pair weights rather than using equal pairwise votes.
- Test a global ranking/meta-ranker in addition to pairwise classifiers.
- Analyze pairwise performance separately for each disease class.
- Inspect oracle-correct/selector-wrong samples to identify useful new features.
- Add entropy, agreement, margin, and validated Phase 3 reliability features only after independent validation.
- Keep the oracle as the upper-bound reference for every future experiment.

13. Phase 4 Status

Core pipeline implemented. Phase 4 has progressed from prediction-feature extraction, through reliability-based selection, to a complete 15-pair dynamic selection framework. The latest corrected benchmark is 78.74% accuracy for pairwise voting versus 91.34% oracle accuracy. The next goal is to close this oracle gap through calibration, better aggregation, class-aware analysis, and improved feature quality.